

Kushin Mukherjee

Postdoctoral Scholar

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Appointments

2025— *Postdoctoral Scholar*, Psychology, Stanford University
Advisor: Judith E. Fan

Education

2019–2025 *PhD*, Psychology, University of Wisconsin-Madison
Advisors: Karen B. Schloss, Timothy T. Rogers
Committee: Michael B. Gleicher, Robert D. Hawkins
2015–2019 *AB*, Cognitive Science and Japanese, Minor in Mathematics, Vassar College
Major advisor: Ken R. Livingston | Thesis advisor: Joshua R. de Leeuw
General Honors
Departmental Honors in Cognitive Science and Japanese

Grants, Honors, & Awards

2021, 2025 Best Paper Honorable Mention, *IEEE VIS*
2025 Computational Modeling Prize in Applied Cognition, *Cognitive Science Society*
2025 Best Paper Award, *ICLR Workshop on Bidirectional Human-AI Alignment*
2025 Student Research Grants Competition Travel Award, *UW-Madison*
2025 Distinguished Paper Award, *McPherson Eye Research Institute, UW-Madison*
2021–2025 Hertz Travel Award, Department of Psychology, *UW-Madison*
2021 Kenzi Valentyn Vision Research Award, *McPherson Eye Research Institute, UW-Madison*
2021 Elsevier Travel Award, *Vision Sciences Society*
2020–2022 Marie Christine Kohler Fellow, *Wisconsin Institute for Discovery, UW-Madison*
2019 Yin-Lien C. Chin Prize for best senior project in Chinese or Japanese, *Vassar College*
2019 Phi Beta Kappa, *Vassar College*
2019 Sigma Xi, *Vassar College*
2018 Psi Chi, *Vassar College*
2016 Summer Program Scholarship, *Ochanomizu University*
2016 Student Exchange Support Scholarship, *Japan Student Service Organization (JASSO)*
2015–2019 Sarah Tod Fitz Randolph Scholarship Fund, *Vassar College*

Working Papers & Preprints

Mukherjee, K., Kim, N. Y., Wehrheim, M., Adolphs, R., & Kar, K. (*under review*). AI-guided stimuli discovery and generation to optimize facial emotion perception studies in autism. *Nature Mental Health*.

Verma, A., **Mukherjee, K.**, Potts, C., Kreiss, E. & Fan, J. (*under review*). CHART-6: Human-centered evaluation of data visualization understanding in vision-language models. *Scientific Reports*.

Tomz, D., **Mukherjee, K.**, Wisher, I., Wulff, F., Braedder, L., Yan, C., Tversky, B., Fusaroli, R., Tylén, K., and Fan, J. (*under revision*). Identifying reliable sources of visual variation in European Paleolithic animal paintings. Identifying the visual features of European Paleolithic cave art that are diagnostic of object category and time period.

Chinni, A. L., Rahman, M. D., Aseniero, B. A., Isenberg, P., **Mukherjee, K.**, Quadri, G. J., Rosen, P., Schloss, K. B., & Weiskopf, D. (*under review*). Guidelines are not rules: Characterizing terminologies around visualization design guidelines.

Ma, W. A., Yue, A., **Mukherjee, K.**, Loeb, S., Demszky*, D., & Fan*, J. (*under revision*). When talk is not enough: Understanding variation in digital whiteboard use in online math tutoring.

Prystawski, B., **Mukherjee, K.**, Wurgaft, D., Nasvytis, L., Li, M. Y., Goodman, N. D., Frank, M. C. (*under review*). AUTO-PSYCH: Automating the science of mind using agent-driven theory discovery and experimentation. <https://arxiv.org/abs/2606.26460>

Peer-reviewed Archival Publications

Asterisks () indicate shared first or senior authorship.*

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| in press | Mukherjee, K. , Rogers, T. T., Lessard, L., Gleicher, M., & Schloss, K. B. (<i>in press</i>). Color-concept associations reveal a new conceptual space. <i>Proceedings of the National Academy of Sciences</i> . |
| in press | Schloss, K. B., Braun, H., Mukherjee, K. , Chinni, A. & Gorelik, S. R. (<i>under revision</i>). Predicting affective connotation of visualizations from their constituent colors. <i>IEEE Transactions on Visualization and Computer Graphics</i> . |
| 2026 | Studdiford, Z., Rogers, T. T., Suresh, S.*, & Mukherjee, K.* (2026). Evaluating steering techniques using human similarity judgments. <i>Proceedings of the Conference on Language Modeling</i> . |
| 2026 | Studdiford, Z., Rogers, T. T., Mukherjee, K.* , & Suresh, S.* (2026). Uncovering the computational ingredients of human-like representations in large language models. <i>Proceedings of the Conference on Language Modeling</i> . |
| 2026 | Mukherjee, K. , Mohapatra, A., Rogers, T. T., & Schloss, K. B. (2026). Large language models estimate fine-grained human color-concept associations. <i>Cognitive Science</i> . |

- 2026 Mukherjee, K., Huey, H., Hebart, M. N., Fan, J. E., & Bainbridge, W. A. (2026). Drawings-of-THINGS: A large-scale dataset containing human sketches of 1,854 object concepts. *Behavior Research Methods*.
- 2026 Chen, A., Kim, S. Y., Franyutti, A., Dharmasiri, A., Mukherjee, K., Fan, J. E., & Russakovsky, O. (2026). Presenting large language models as companions affects what mental capacities people attribute to them. *Proceedings of the CHI Conference on Human Factors in Computing Systems*.
- 2025 Mukherjee, K., Ren, D., Moritz, D., & Assogba, Y. (2025). EncQA: Evaluating visual encoding understanding for visualizations in large vision-language models. *IEEE Transactions on Visualization and Computer Graphics*.
- 2025 Braun, H., Mukherjee, K., Gorelik, S. R. & Schloss, K. B. (2025). Affective color scales for colormap data visualizations. *IEEE Transactions on Visualization and Computer Graphics*. ***Best Paper Honorable Mention Award**
- 2025 Suresh, S.*, Mukherjee, K.*, Giallanza, T., Yu, X., Patil, M., Cohen, J. D., & Rogers, T. T. (2025). AI-Enhanced Semantic Feature Norms for 786 Concepts. *TopiCS in Cognitive Science*.
- 2025 Mukherjee, K., & Rogers, T. T. (2025). Using drawings and deep neural networks to characterize the building blocks of human visual similarity. *Memory & Cognition*.
- 2023 Suresh, S., Mukherjee, K., Yu, X., Huang, W., Padua, L., & Rogers, T. T. (2023). Conceptual structure coheres in human cognition but not in large language models. *Proceedings of the Conference on Empirical Methods in Natural Language Processing (EMNLP)*.
- 2023 Mukherjee, K., Lu, X., Huey, H., Vinker, Y., Shamir, A., & Fan, J. E. (2023). SEVA: Leveraging sketches to evaluate alignment between human and machine visual abstraction. *Advances in Neural Information Processing Systems (NeurIPS), Datasets & Benchmarks Track*.
- 2021 Mukherjee, K., Yin, B., Sherman B. E., Lessard, L. & Schloss, K. B. (2021). Context matters: Semantic discriminability theory for perceptual encoding systems. *IEEE Transactions on Visualization and Computer Graphics*. ***Best Paper Honorable Mention Award**

Peer-reviewed Proceedings & Workshop Papers

- 2025 Suresh, S.*, Mukherjee, K.*, Giallanza, T., Yu, X., Patil, M., Cohen, J. D., & Rogers, T. T. (2025). AI-assisted semantic norms for 786 concepts. *ICLR Workshop on Bidirectional Human-AI Alignment*. ***Best Paper Award**
- 2025 Suresh, S.*, Mukherjee, K.*, Giallanza, T., Yu, X., Patil, M., Cohen, J. D., & Rogers, T. T. (2025). AI-assisted semantic norms for 786 concepts. *Proceedings of the 47th Annual Meeting of the Cognitive Science Society*. ***Modeling Prize in Applied Cognition**
- 2024 Mukherjee, K. (2024). Shaping vision through drawing. *Nature Reviews Psychology*.

- 2024 Verma, A., **Mukherjee, K.**, Kreiss, E., Potts, C., and Fan, J. (2024). Evaluating human and machine understanding of data visualizations. *Proceedings of the 46th Annual Meeting of the Cognitive Science Society*.
- 2024 **Mukherjee, K.**, Suresh, S., Yu, X., & Lupyan, G. (2024). The role of shared labels and shared experiences in representational alignment. *International Conference on Learning Representations (ICLR) Re-Align Workshop*.
- 2024 Suresh, S., Huang, W., **Mukherjee, K.**, & Rogers, T.T. (2024). Categories vs semantic features: What shapes the similarities people discern in photographs of objects? *International Conference on Learning Representations (ICLR) Re-Align Workshop*.
- 2023 **Mukherjee, K.**, Kim, N. Y., Alamooti, S. T., Adolphs, R., & Kar, K. (2023). Leveraging Artificial Neural Networks to Enhance Diagnostic Efficiency in Autism Spectrum Disorder: A Study on Facial Emotion Recognition. *Conference on Cognitive Computational Neuroscience*.
- 2023 **Mukherjee, K.**, Lu, X., Huey, H., Vinker, Y., Shamir, A., & Fan, J. E. (2023). Evaluating machine comprehension of sketch meaning at different levels of abstraction. *Proceedings of the 45th Annual Meeting of the Cognitive Science Society*.
- 2023 Suresh, S., **Mukherjee, K.** & Rogers, T. T. (2023). Semantic Feature Verification in FLAN-T5. *International Conference on Learning Representations (ICLR), Tiny Papers Track*.
- 2023 **Mukherjee, K.**, Suresh, S. & Rogers, T. T. (2023). Human-machine cooperation for semantic feature generation. *International Conference on Learning Representations (ICLR), Tiny Papers Track*.
- 2020 **Mukherjee, K.**, & Rogers, T. T. (2020). How does task structure shape representations in deep neural networks? *2nd NeurIPS Workshop on Shared Visual Representations in Human and Machine Intelligence*.
- 2019 **Mukherjee, K.**, Hawkins, R. D., & Fan, J. E. (2019). Communicating semantic part information in drawings. *Proceedings of the 41st Annual Meeting of the Cognitive Science Society*.

Book Chapters

- 2024 Schloss, K. B., Schoenlein, M. A., & **Mukherjee, K.** (2024). Color semantics for visual communication. In R. B. D. A. Szafir, M. Chen, D. J. Edward, B. Fisher, & L. M. Padilla (Ed.), *Visualization Psychology*.

Conference Presentations

- 2026 Tomz, D., **Mukherjee, K.**, Wisher, I., Wulff, F., Braedder, L., Yan, C., Tversky, B., Fusaroli, R., Tylén, K., & Fan, J. E. (2026). Identifying the visual features of European Paleolithic cave art that are diagnostic of object category and time period. Poster presented at the 48th Annual Meeting of the Cognitive Science Society.
- 2026 Caren, M., **Mukherjee, K.**, Agrawala, M., & Fan, J. E. (2026). Generating acoustic signals to achieve both referential and aesthetic goals. Poster presented at the 48th Annual Meeting of the Cognitive Science Society.

- 2026 Li, Z., Colón, Y. I., **Mukherjee, K.**, & Rogers, T. T. (2026). Are Faces, Places, and Objects Encoded in the Same Locations across Individual Brains? Poster presented at the 48th Annual Meeting of the Cognitive Science Society.
- 2026 **Mukherjee, K.** & Fan, J. E. (2026). Designers choose to show data visualizations that observers can reliably interpret. Poster presented at the 26th Annual Meeting of the Vision Sciences Society.
- 2026 Tomz, D., **Mukherjee, K.**, Wisher, I., Wulff, F., Braedder, L., Yan, C., Tversky, B., Fusaroli, R., Tylén, K., & Fan, J. E. (2026). Identifying the visual features of European Paleolithic cave paintings that are diagnostic of category, age, and location. Poster presented at the 26th Annual Meeting of the Vision Sciences Society. **Vision Sciences Society Travel Award*
- 2026 Chinni, A., **Mukherjee, K.** & Schloss, K. B. (2026). Semantic distance predicts ability to interpret texture meaning in information visualizations. Poster presented at the 26th Annual Meeting of the Vision Sciences Society.
- 2026 Braun, H., **Mukherjee, K.**, Gorelik, S. R., & Schloss, K. B. (2026). Predicting the affective connotation of colormap data visualizations from estimated emotional associations of constituent colors. Poster presented at the 26th Annual Meeting of the Vision Sciences Society.
- 2025 **Mukherjee, K.**, Ren, D., Moritz, D. & Assogba, Y. (2025). EncQA: Evaluating visual encoding understanding for visualizations in large vision-language models. Talk presented at IEEE VIS 2025.
- 2025 Braun, H., **Mukherjee, K.**, Gorelik, S. R. & Schloss, K. B. (2025). Affective color scales for colormap data visualizations. Talk presented at IEEE VIS 2025.
- 2025 Chinni, A., Colón, Y. I., **Mukherjee, K.**, Gleicher, M. L., & Schloss, K. B. (2025). Development of the NBT-53 Texture Library for the Study of Texture Semantics. Poster presented at the 66th Annual Meeting of the Psychonomic Society.
- 2025 Braun, H., **Mukherjee, K.**, Gorelik, S. R. & Schloss, K. B. (2025). Affective color scales for colormap data visualizations. Poster presented at the 25th Annual Meeting of the Vision Sciences Society.
- 2025 Chinni, A., Colón, Y. I., **Mukherjee, K.**, Gleicher, M. L., & Schloss, K. B. (2025). Development of the NBT-53 Texture Library for the Study of Texture Semantics. Poster presented at the 25th Annual Meeting of the Vision Sciences Society.
- 2025 Suresh, S. **Mukherjee, K.**, Giallanza, T., Yu, X., Patil, M., Cohen, J. D., & Rogers, T. T. (2025). AI-assisted semantic norms for 786 concepts. Talk presented at the 47th Annual Meeting of the Cognitive Science Society.
- 2025 Li, Z., Colón, Y. I., **Mukherjee, K.**, Rogers, T. T. (2025). Iterated LASSO reveals highly distributed and variable representations of faces, places, and objects. Poster presented at the 47th Annual Meeting of the Cognitive Science Society.
- 2025 Suresh, S. **Mukherjee, K.**, Giallanza, T., Yu, X., Patil, M., Cohen, J. D., & Rogers, T. T. (2025). AI-assisted semantic norms for 786 concepts. Talk and Poster presented at the Bi-Align Workshop at ICLR.
- 2024 **Mukherjee, K.**, Rogers, T. T., & Schloss, K. B. (2024). Estimating human color-concept associations from multimodal language models. Poster presented at the 46th Annual Meeting of the Cognitive Science Society.

- 2024 Verma, A., **Mukherjee, K.**, Kreiss, E., Potts, C., and Fan, J. (2024). Evaluating human and machine understanding of data visualizations. Poster presented at the 46th Annual Meeting of the Cognitive Science Society.
- 2024 Suresh, S., **Mukherjee, K.**, & Rogers, T. T. (2024). Can deep convolutional networks explain the semantic structure that humans see in photographs? Poster presented at the 46th Annual Meeting of the Cognitive Science Society.
- 2023 **Mukherjee, K.**, Kim, N. Y., Alamooti, S. T., Adolphs, R., & Kar, K.. (2023). Leveraging Artificial Neural Networks to Enhance Diagnostic Efficiency in Autism Spectrum Disorder: A Study on Facial Emotion Recognition. Talk and Poster presented at the Conference on Cognitive Computational Neuroscience.
- 2023 **Mukherjee, K.**, Lessard, L., & Schloss K. B. (2023). How do people map colors to concepts? Modeling assignment inference as evidence accumulation. Talk presented at the 23rd Annual Meeting of the Vision Sciences Society.
- 2023 Suresh, S., **Mukherjee, K.**, & Rogers T. T. (2023). Can deep convolutional networks explain the semantic structure that humans see in photographs? Talk presented at the 23rd Annual Meeting of the Vision Sciences Society.
- 2023 Fan, J. E., **Mukherjee, K.**, Huey, H., Hebart, M., & Bainbridge, W. (2023). THINGS-drawings: A large-scale dataset containing human sketches of 1,854 object concepts. Talk presented at the 23rd Annual Meeting of the Vision Sciences Society.
- 2023 **Mukherjee, K.**, Lu, X., Huey, H., Vinker, Y., Shamir, A., & Fan, J. E. (2023). Evaluating machine comprehension of sketch meaning at different levels of abstraction. Poster presented at the 23rd Annual Meeting of the Vision Sciences Society.
- 2022 Armendariz, M., **Mukherjee, K.**, Shang, J., & Kar, K. (2022). Probing the functional relevance of side-reads and bypass-connections in the primate ventral stream during visual object recognition using deep neural networks. Poster presented at the 22nd Annual Meeting of the Vision Sciences Society.
- 2022 **Mukherjee, K.**, Schloss, K. B., Lessard, L., Gleicher, M., & Rogers, T.T. (2022). Color-concept associations reveal an abstract conceptual space. Poster presented at the 22nd Annual Meeting of the Vision Sciences Society.
- 2021 **Mukherjee, K.**, Rogers, T.T., Lessard, L., Gleicher, M., & Schloss, K. B. (2021). Mapping a low-dimensional space of color-concept associations. Poster presented at the 21st Annual Meeting of the Vision Sciences Society. **Vision Sciences Society Travel Award*
- 2021 **Mukherjee, K.**, Yin, B., Sherman B. E., Lessard, L. & Schloss, K. B. (2021). Context matters: Semantic discriminability theory for perceptual encoding systems. Talk presented at the 62nd Annual Meeting of the Psychonomic Society.
- 2021 **Mukherjee, K.**, Yin, B., Sherman B. E., Lessard, L. & Schloss, K. B. (2021). Context matters: Semantic discriminability theory for perceptual encoding systems. Talk presented at IEEE VIS 2021.
- 2020 **Mukherjee, K.**, & Rogers, T. T. (2020). How does task structure shape representations in deep neural networks? Poster presented at the 2nd NeurIPS Workshop on Shared Visual Representations in Human and Machine Intelligence.
- 2020 **Mukherjee, K.**, & Rogers, T. T. (2020). Finding meaning in simple sketches: How do humans and deep networks compare? Poster presented at the 20th Annual Meeting of the Vision Sciences Society.

- 2019 Mukherjee, K., Hawkins, R. D., & Fan, J. E. (2019). Communicating semantic part information in drawings. Poster presented at the 41st Annual Meeting of the Cognitive Science Society.

Invited Talks & Seminars

- 2026 Designers choose to show data visualizations that observers can reliably interpret. *Vis X Vision Workshop @ VSS 2026*.
- 2026 Using drawings and data visualizations to express knowledge. *UIUC Attention & Perception Seminar*.
- 2026 Using drawings and data visualizations to express knowledge. *Stanford FriSem*.
- 2025 Cognitive abstractions for visual communication. *McPherson Eye Research Institute Seminar*.
- 2025 What makes different graphs effective for answering different questions? *Nightingale Labs, Stanford University*.
- 2025 Perceptual semantics for visual and haptic communication. *Wisconsin Institute for Discovery Seminar Series*.
- 2024 EncQA: Evaluating visual encoding understanding for visualizations in large vision-language models. *Apple Human Centered Machine Intelligence Sync*.
- 2023 Using drawings to understand human semantic cognition. *MRC Cognition and Brain Sciences Unit, University of Cambridge*.
- 2023 THINGS-drawings: A large-scale dataset containing human sketches of 1,854 object concepts. *Cognitive Brown Bag, University of Chicago*.
- 2023 THINGS-drawings: A large-scale dataset containing human sketches of 1,854 object concepts. *Cognitive Tools Lab, UC San Diego*.
- 2023 Evaluating machine comprehension of sketch meaning at different levels of abstraction. *Stanford NeuroAI Lab, Stanford University*.
- 2023 Matrix completion techniques for behavioral sciences. *AI & Society Seminar, UW-Madison*.
- 2023 Using drawings and deep neural networks to characterize the building blocks of human visual similarity. *Wisconsin Institute for Discovery Seminar Series*.
- 2022 Using line drawings to understand what deep learning models see. *McPherson Eye Research Institute Seminar*

Internships & Work Experience

- 06 - 09, 2024 AI/ML Intern, *Apple*
- 08, 2021 Summer School Fellow, *MIT Center for Minds, Brains, and Machines*
- 06 - 08, 2018 CSLI Summer Intern, *Stanford University*

Teaching

GUEST INSTRUCTOR, *STANFORD UNIVERSITY*

- 2026 PSYCH 230, *Computational Models of the Creative Process*

GRADUATE TEACHING ASSISTANT, *UNIVERSITY OF WISCONSIN-MADISON*

2022 PSYCH 454, *Behavioral Neuroscience*
2021 PSYCH 210, *Statistics for Psychology*
2020 PSYCH 414, *Cognitive Psychology*

UNDERGRADUATE TEACHING ASSISTANT, *VASSAR COLLEGE*

2017 COGS 211, *Perception and Action*

Advising

UNDERGRADUATE STUDENTS

2025 - Avery Yue (Stanford)
2025 - David Tomz (Stanford)
2025 - 2026 Zach Studdiford (UW-Madison)
2024 - 2025 Rainy Jin (UW-Madison)
2024 - 2026 Halle Braun (UW-Madison)
2024 - 2025 Ria Pattekar (UW-Madison)
2024 - 2025 Ankit Mohapatra (UW-Madison)
2023 - 2024 Nancy Davis (UW-Madison)
2022 - 2023 Jonah Manaligold (UW-Madison)
2022 - 2023 Janani Sundar (UW-Madison)
2022 - 2023 Rio Aguina-Kang (UCSD)
2022 Lisa Padua (Albany State)
2020 - 2021 Brianne E. Sherman (UW-Madison)

Professional Service

WORKSHOP ORGANIZATION

2026 ReAlign: ICLR Workshop on Representational Alignment, *14th International Conference on Learning Representations*
2024 COGGRAPH: Building bridges between cognitive science and computer graphics, *46th Annual Meeting of the Cognitive Science Society*
2022 Images2Symbols: Drawing as a Window into the Mind, *44th Annual Meeting of the Cognitive Science Society*

PROGRAM COMMITTEES

2025, 2026 VIS×AI: Workshop on Visualization for AI Explainability, *IEEE VIS*

MENTORING PROGRAMS

2025 Stanford Psychology *Paths to PhD*
2024-2025 Cognitive Science Society Mentorship Program
2024 UW-Madison Psychology PhD Mentorship Panel

AD HOC REVIEWING

Journals & Books

Nature, Cognition, Open Mind, Communications Biology, Nature Reviews Psychology, Transactions on Visualization and Computer Graphics (TVCG), EPJ Data Science, Visualization Psychology

Conference Proceedings and Workshops

Neural Information Processing Systems (NeurIPS), Association for Computational Linguistics (ACL), International Conference on Learning Representations (ICLR), Conference on Language Modeling (COLM), Conference on Computational Cognitive Neuroscience (CCN), IEEE VIS, ACM Conference on Human Factors in Computing Systems (CHI), SIGGRAPH, Annual Meeting of the Cognitive Science Society (CogSci), NeurIPS Workshop on Shared Visual Representations in Humans and Machines (SVRHM), ICLR Workshop on Representational Alignment (ReAlign)

DEPARTMENTAL SERVICE

2020-2022	University of Wisconsin-Madison Psychology Colloquium Committee
2017-2019	Vassar College Cognitive Science Majors' Committee, <i>Chair</i>
2016-2017	Vassar College Student Association Finance Committee

AFFILIATIONS

2019-	Cognitive Science Society
2020-	Vision Sciences Society
2021-2022	Psychonomics Society

References

Dr. Timothy T. Rogers | University of Wisconsin-Madison, Department of Psychology
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Dr. Karen B. Schloss | University of Wisconsin-Madison, Department of Psychology
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Madison, WI 53706-1611
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Dr. Judith E. Fan | Stanford University, Department of Psychology
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